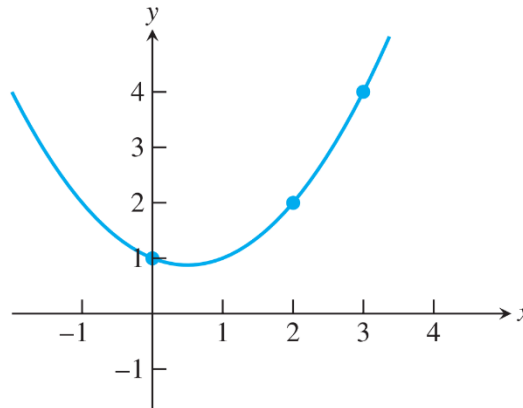


Topic Covered:

- Data and interpolating functions
- Least squares and the normal equations

1. Find an interpolating polynomial for the data points (0,1), (2,2), and (3,4) in the following figure. Using Lagrange's formula



Answer:

Lagrange Interpolation Formula:

For $n + 1$ points (x_i, y_i) , the interpolating polynomial of degree n is:

$$P_n(x) = \sum_{i=0}^n y_i l_i(x), l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

Points = 3 $\rightarrow n = 2$. So, the interpolating polynomial of degree n is

$$P_2(x) = 1 \cdot l_0(x) + 2 \cdot l_1(x) + 4 \cdot l_2(x)$$

$$P_2(x) = 1 \cdot \frac{(x-2)(x-3)}{(0-2)(0-3)} + 2 \cdot \frac{(x-0)(x-3)}{(2-0)(2-3)} + 4 \cdot \frac{(x-0)(x-2)}{(3-0)(3-2)}$$

$$P_2(x) = 1 \cdot \frac{(x-2)(x-3)}{6} - 2 \cdot \frac{x(x-3)}{2} + 4 \cdot \frac{x(x-2)}{3}$$

$$P_2(x) = \frac{(x-2)(x-3) - 6x(x-3) + 8x(x-2)}{6}$$

$$P_2(x) = \frac{3x^2 - 3x + 6}{6} = \frac{1}{2}x^2 - \frac{1}{2}x + 1$$

Verification: $P_2(0) = 1, P_2(2) = 2, P_2(3) = 4$ ✓

$$\Rightarrow P_2(x) = \frac{1}{2}x^2 - \frac{1}{2}x + 1$$



2. Solve the least squares problem using normal equations $A^T A x = A^T b$.

$$\begin{bmatrix} 1 & -4 \\ 2 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3 \\ 15 \\ 9 \end{bmatrix}$$

Also find the residual vector and its Euclidean norm.

Answer:

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} -3 \\ 15 \\ 9 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 9 & 6 \\ 6 & 29 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 45 \\ 75 \end{bmatrix}$$

$$A^T A x = A^T b \Rightarrow \begin{bmatrix} 9 & 6 \\ 6 & 29 \end{bmatrix} x = \begin{bmatrix} 45 \\ 75 \end{bmatrix}$$

Solve $A^T A x = A^T b \rightarrow$ Gaussian elimination yields: $x = \begin{bmatrix} \frac{19}{5} \\ \frac{9}{5} \end{bmatrix}$. [the best line fits the points is $y = \frac{19}{5}t + \frac{9}{5}$]

Now,

$$Ax = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} \frac{19}{5} \\ \frac{9}{5} \end{bmatrix} = \begin{bmatrix} -3.4 \\ 13 \\ 11.2 \end{bmatrix}$$

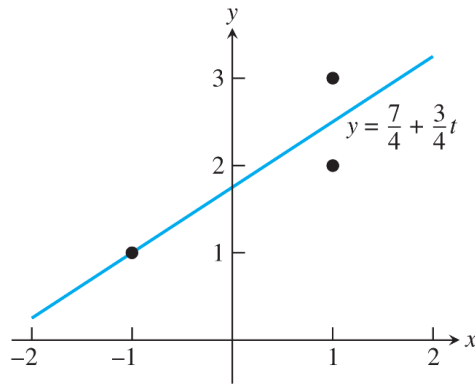
$$\text{Residual: } r = b - Ax = \begin{bmatrix} -3 \\ 15 \\ 9 \end{bmatrix} - \begin{bmatrix} -3.4 \\ 13 \\ 11.2 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 2 \\ -2.2 \end{bmatrix}.$$

$$\text{Euclidean norm: } \|r\|_2 = \sqrt{\sum r_i^2} = \sqrt{\left(\frac{2}{5}\right)^2 + 2^2 + \left(-\frac{11}{5}\right)^2} = \sqrt{9} = 3.$$

Orthogonality check: $A^T r = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ✓ [Note: $A^T A x = A^T b \Rightarrow A^T (b - Ax) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$]



3. Find the line that best fits the three data points $(t, y) = (1, 2), (-1, 1),$ and $(1, 3)$ in following figure.



Answer:

Rule: Linear Least Squares Fitting

To find the best line $y = mt + c$ for the three data points, use the form:

$$\begin{bmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}, \quad \text{as } Ax = b$$

Then solve: $A^T Ax = A^T b$, to obtain the best m and c . Therefore, the best line is $y = mt + c$.

Now, the points $(t_i, y_i) = (1, 2), (-1, 1), (1, 3)$, are substituted into the model $y = mt + c$, yielding:

$$2 = m(1) + c,$$

$$1 = m(-1) + c,$$

$$3 = m(1) + c,$$

In matrix form:

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}.$$

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$A^T Ax = A^T b \Rightarrow \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} x = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

$$\text{Solve } A^T Ax = A^T b \rightarrow \text{Gaussian elimination gives: } x = \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ \frac{7}{4} \end{bmatrix}.$$

$$\text{Thus, the best line that fits the three data points is: } y = \frac{3}{4}t + \frac{7}{4}.$$

Verification:

$$\text{At } t = 1: y = 2.5 \quad \checkmark \text{ (average of 2 and 3)}$$

$$\text{At } t = -1: y = 1 \quad \checkmark$$

